

Solution: Let, $x = 65^{1000} - 8^{2001} + 3^{177}$, $y = 79^{1212} - 9^{2392} + 2^{2001}$ and

$$z = 24^{4493} - 5^{8192} + 7^{1777}$$

Suppose, for the sake of contradiction, each of the three products xy , xz and yz are negative.

$\therefore xy$ is negative, exactly one of x or y must be negative.

WLOG, let x be negative and y be positive

$\therefore xz$ is negative, $\therefore z$ is positive

$\therefore$ Both y and z are ~~negative~~ ^{positive}, yz is positive. This is a contradiction to the fact that yz is ^{negative}.

$\therefore$ At least one of the products xy , xz or yz must be non-negative. $\square$

15. Prove or disprove that there is a rational number x and an irrational number y such that x^y is irrational. (Very Good Prob)

Solution: Let, $x = 2$ and $y = (\sqrt{2} + \frac{1}{2})^x = \frac{2}{1}$. By Defn. 2, x is rational

~~From Sec 1.7, Example 11, we know y is irrational.~~
Let, y be rational. By Defn. 2, $\exists$ integers a and b in lowest terms where $b \neq 0$ such that $(\sqrt{2} + \frac{1}{2}) = \frac{a}{b} \Rightarrow \sqrt{2} = \frac{a}{b} - \frac{1}{2} = \frac{2a-b}{2b}$. It's easy to see

that $2a-b$ and $2b$ are integers. $\therefore b \neq 0 \therefore 2b \neq 0$. By Defn. 2, $\sqrt{2}$ is rational

This contradicts the fact that $\sqrt{2}$ is irrational from Sec 1.7, Example 11.

$\therefore y = (\sqrt{2} + \frac{1}{2})$ is irrational. $x^y = 2^{(\sqrt{2} + \frac{1}{2})}$. If it is irrational, then we are done. Else, new $x = 2^{(\sqrt{2} + \frac{1}{2})}$, new $y = (\sqrt{2} - \frac{1}{2})$

Let, new y be rational. By Defn. 2, $\exists$ integers c and d in lowest terms such that $d \neq 0$ and $\sqrt{2} - \frac{1}{2} = \frac{c}{d} \Rightarrow \sqrt{2} = \frac{c}{d} + \frac{1}{2} = \frac{2c+d}{2d}$. It's easy to see that both $2c+d$ and $2d$ are integers. $\therefore d \neq 0, \therefore 2d \neq 0$. By Defn. 2, $\sqrt{2}$ is rational. This contradicts the fact $\sqrt{2}$ is irrational from Sec 1.7, Example 11.